Retro Météo

A SWISS CLIMATE EXPLORER

Visualising historic weather-station data,
from last month to more than a century ago.

PROCESS BOOK

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COM-480

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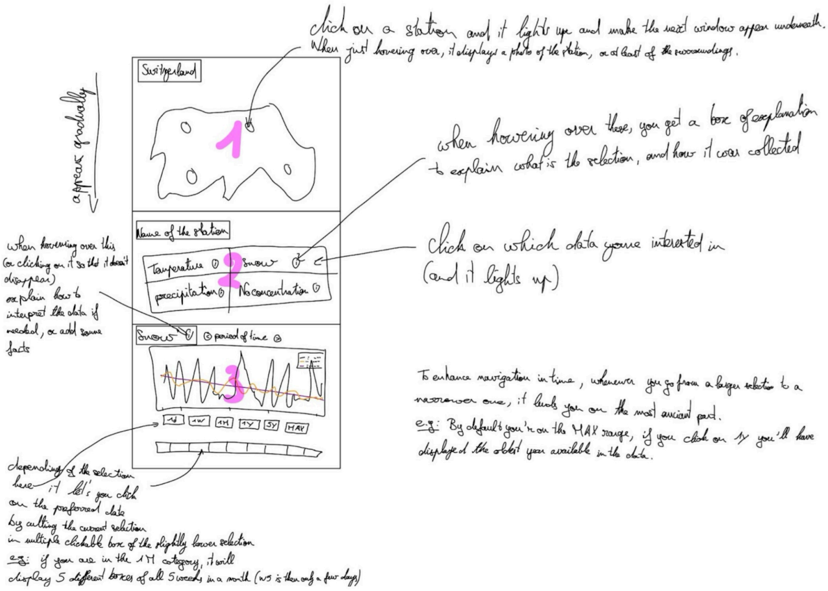
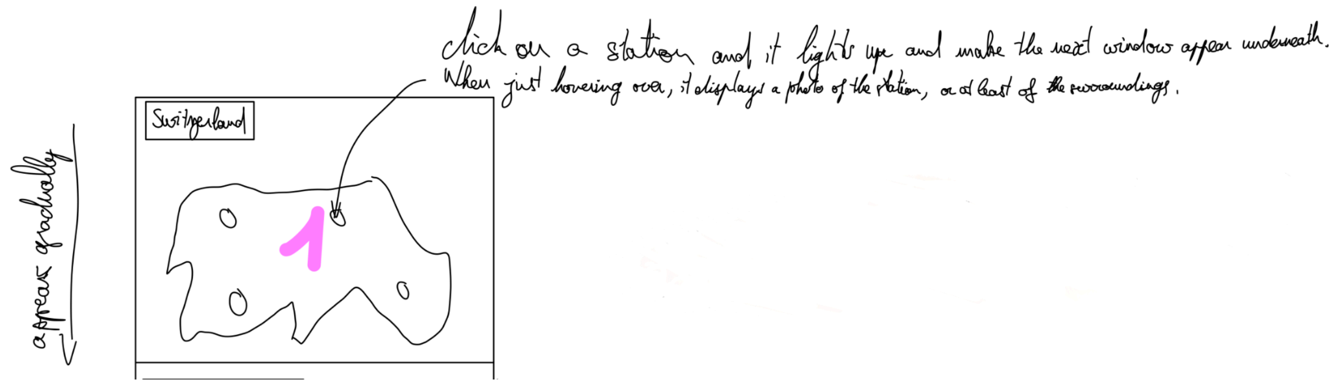
Remember the first sketches?



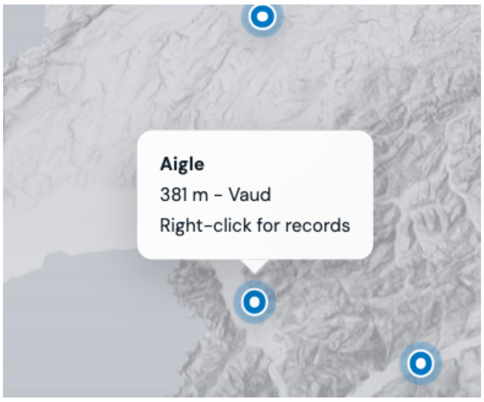
This was the sketch from Milestone 2

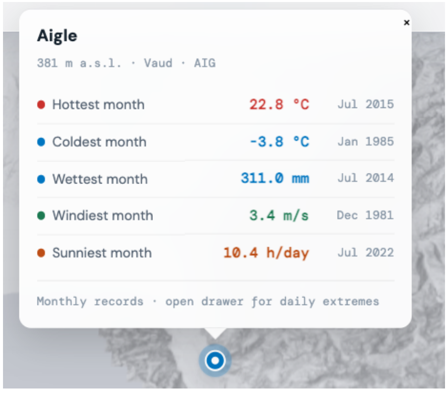
Let's dive deeper into each planned feature, check how we've implemented it, and see where we faced challenges.

Screen 1: Map



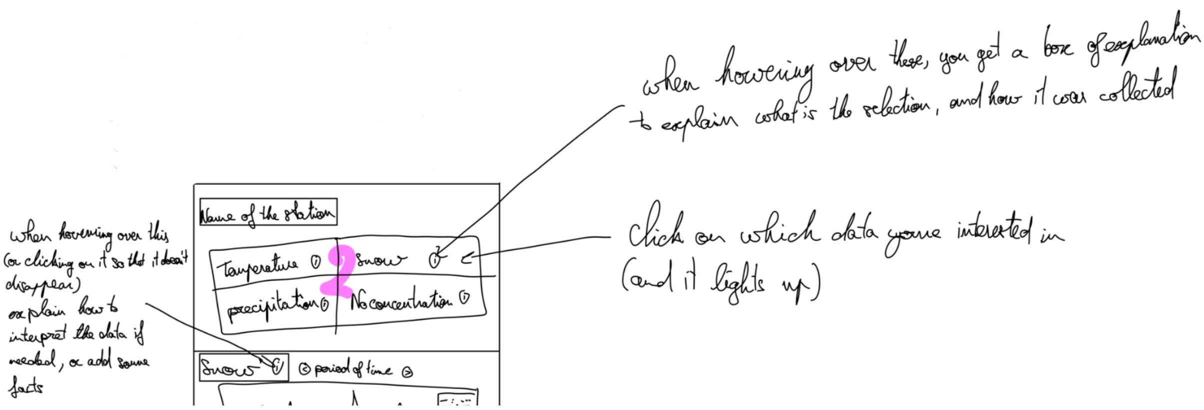
Well, indeed we've implemented the map with all the little stations you can click on, and hover on; the main change is that we did not add the photos, because they were looking a bit out of place with our minimalist theme, but the map is a 3D map, so when zooming you can see the surrounding mountains, and it gives an idea of the type of environment.





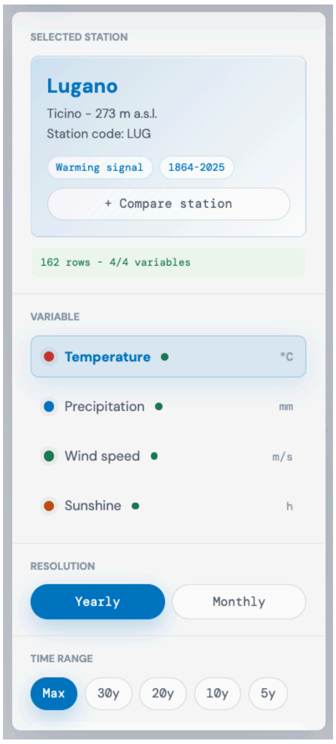
Another feature we've changed is the gradual appearing of the elements. It could have made sense for a phone only usage, but for a big screen it is more annoying than anything. We tried to implement it, but the problem is that there is a conflict between zooming on the map, and scrolling down to do the selection of your feature of interest, which makes the website a frustration generator. To avoid this issue, we've decided to have one screen with everything (*One screen to rule them all*), and the only scrollable part is the selection panel on the left of the screen.

Screen 2: Selection panel



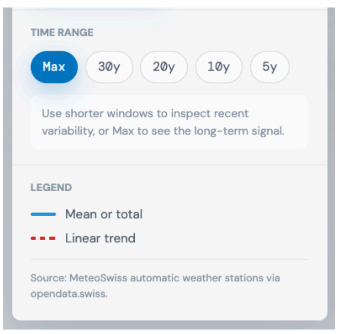
As stated in the previous part, the selection panel is now on the right. Initially, we only thought about a very minimalist version with 4 features and some more information when hovering over each (i), but we realised that we needed to add the part that handles the resolution and time range selection on it to make the last screen more readable.

The initial time range and resolution were supposed to be in the third screen



The final selection panel looks more now, like a control table for close to all the possible features available, and especially for one that we're particularly proud of, which is the comparison.
It may look like a simple feature to implement, but it clearly isn't, and we had to go through a lot of trial and error to find a way to make the comparison intuitive and easily comprehensive.
The solution we opted for is that the button appear whenever you've clicked on a station, and then when clicking on the [+ Compare station] the button changes for:

You can then choose whatever station you want to compare, and even the same station, so that you can see the evolution of weather very precisely.



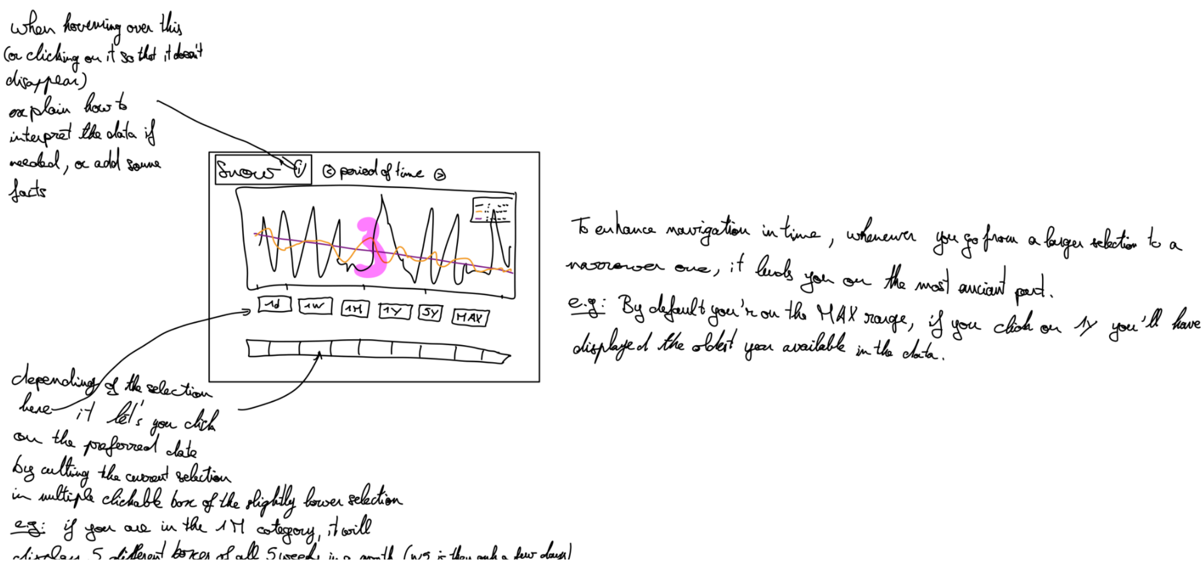
Onto the next challenge: the time resolution.
we had to find the right balance between an interesting resolution to get insights from the data, but also a not too demanding display of the data (especially when going for the Max range if we had a day to day resolution) hence the absence of a more granular option than monthly. It also makes sense from a climate analysis point of view, since the day to day weather doesn't really come with useful information about elevation of temperature, and mean over months make way more sense in that regard.

Finally, the last tricky implementation of this screen is the time range. If you select another time range than the Max:

You have the possibility to navigate until you reach the oldest or newest data. This was difficult to make sure that it is coherent with the current looked station, since all the stations have a different time of first data. For some it's in the 60's, for others it's also in the 60's, but not 1960, 1860!

So for our implementation to work and make sense for all ranges of measurements for all stations, we had to check manually the csv's of all the 16 stations for all four variables, and find the middle way so that the selection panel stays relevant for both very old and very new stations. We're very happy with the current options!

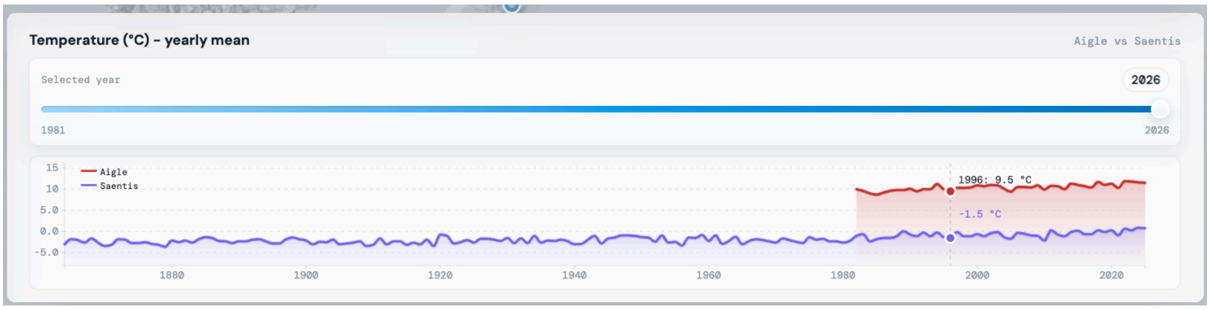
Screen 3: Data & insights



We're getting to the toughest part to design, how to make the data insightful and not soporific?

The first choice was to put all the selection part in the selection panel, so that when you look at this screen, you stay focused on the graphs and facts. As we had planned, graphs alone had a great chance to be slightly too boring, so we added the Temperature records, along with the trend (it was originally planned to see it only when hovering over an (i) but we realised it's at least as important as the graph, so it should always stay visible.

The second challenge is to be able to compare two stations on a variable, even though they don't have the same span of activity (as stated before), we thought the best way to display this is to let the graph be as long as the station that has the largest span. Here is an example with the comparison of temperature between Aigle and Saentis, our highest station:



Overall

For our finished product we have one screen. We can interact with it but still it's one screen. This is our biggest strength while also being our biggest weakness. The strength is that it makes the website very easy to get in hand, and thus really help make the most out of the dataset we've used. The weakness is that it can get a little boring. To add some interactivity (and fun) and to do all the extra ideas we talked about in milestone 2, we've decided to add two features, which you can click on and close at any time. They're called "Story" and "🏆Hall of Fame". Story is some fun facts about the data we have, and Hall of Fame retraces all the records in our data, and animate the map to go to each designated station, which looks very cool.

Overall this was a very interesting project, and most of all helped us grasp what modern data visualization allows us to do. This project was made with the assistance of generative AI. Like any tool, it supported our process, but the ideas, choices, and craft behind the work are ours.

Peer assessment

For this project, we’ve split up the work following this repartition; Thomas and Noa did the coding and Gaspard did the sketches, and the milestones.